

(12) **United States Patent**
Matsui et al.

(10) **Patent No.:** **US 12,405,418 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **MULTICORE OPTICAL FIBER, METHOD OF DESIGNING MULTICORE OPTICAL FIBER, AND OPTICAL TRANSMISSION METHOD**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/392,976**

(22) Filed: **Dec. 21, 2023**

(65) **Prior Publication Data**

US 2024/0118486 A1 Apr. 11, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/294,892, filed as application No. PCT/JP2019/043826 on Nov. 8, 2019, now Pat. No. 11,899,237.

(30) **Foreign Application Priority Data**

Nov. 21, 2018 (JP) 2018-218153

(51) **Int. Cl.**

G02B 6/02 (2006.01)

G02B 27/00 (2006.01)

H04J 14/00 (2006.01)

(52) **U.S. Cl.**

CPC **G02B 6/02042** (2013.01); **G02B 27/0012** (2013.01); **H04J 14/052** (2023.08)

(58) **Field of Classification Search**

CPC G02B 6/02042
See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,653,852 A * 3/1987 Suzuki G02B 6/4403
385/114

7,079,736 B2 7/2006 Takahashi et al.

(Continued)

FOREIGN PATENT DOCUMENTS

JP 2013088457 A * 5/2013

JP 2014010266 A * 1/2014

(Continued)

OTHER PUBLICATIONS

JP_5660627_B2 English translation (Year: 2015).*

(Continued)

Primary Examiner — Charlie Y Peng

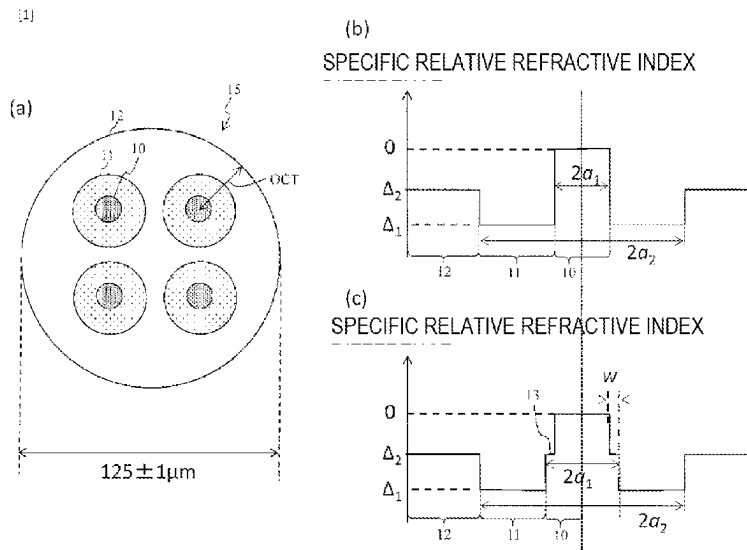
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(57)

ABSTRACT

It is an object of the present invention to provide a multicore optical fiber, a design method for the multicore optical fiber and an optical transmission method using the multicore optical fiber including four cores having a standard cladding diameter of $125 \pm 1 \mu\text{m}$ for an existing single mode optical fiber covering several thousands of kilometers of transmission. The multicore optical fiber according to the present invention disposes two-stage claddings with different refractive indices around each core, and designates as a predetermined range, a core radius a_1 , a radius a_2 of a first cladding region surrounding each core, specific refractive index Δ_1 relative to the core of the first cladding region and a specific refractive index Δ_2 relative to the core of a second cladding region including four cores and the first cladding region.

9 Claims, 14 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

7,844,155	B2	11/2010	Bickham et al.	
8,737,793	B2 *	5/2014	Imamura	G02B 6/0365 385/126
9,081,129	B2	7/2015	Matsuo et al.	
11,366,266	B2	6/2022	Sagae et al.	
2008/0279515	A1	11/2008	Bickham et al.	
2020/0379168	A1	12/2020	Mukasa	
2022/0003920	A1	1/2022	Matsui et al.	

FOREIGN PATENT DOCUMENTS

JP	5660627	B2 *	1/2015
JP	2018021999		2/2018
JP	6560806		8/2019

OTHER PUBLICATIONS

Gonda et al., "125 μ m 5-core fibre with heterogeneous design suitable for migration from singlecore system to multi-core system"

ECOC, 42nd European Conference and Exhibition on Optical Communications, Dusseldorf, Sep. 18-22, 2016, pp. 547-549.

Hayashi et al., "Design and fabrication of ultra-low crosstalk and low-loss multi-core fiber," Opt. Express, Aug. 2011, 19(17):16576-16592.

Matsui et al., "118.5 Tbit/s Transmission over 316 km-Long Multi-Core Fiber with Standard Cladding Diameter," OECC, Aug. 2017, 2 pages.

Matsui et al., "Design of 125 μ m cladding multi-core fiber with full-band compatibility to conventional single-mode fiber," ECOC, Sep. 2015, 3 pages.

Takahashi et al., "First Demonstration of MC-EDFA-Repeatered SDM Transmission of 40 $\times$ 128-Gbit/s PDM-QPSK Signals per Core over 6, 160-km 7-core MCF," ECOC Postdeadline Papers, Th3C3, Sep. 2012, 3 pages.

Tamura et al., "Aeff Expanded Non-Binding 2-Core Fiber Having Cladding Diameter of 125 μ m," IEICE (The Institute of Electronics, Information and Communication Engineers) Society Conference, B-10-2, Sep. 2016, p. 193, 4 pages (with English Translation).

Winzer et al., "Penalties from In-Band Crosstalk for Advanced Optical Modulation Formats," ECOC Technical Digest, Sep. 2011, 3 pages.

* cited by examiner

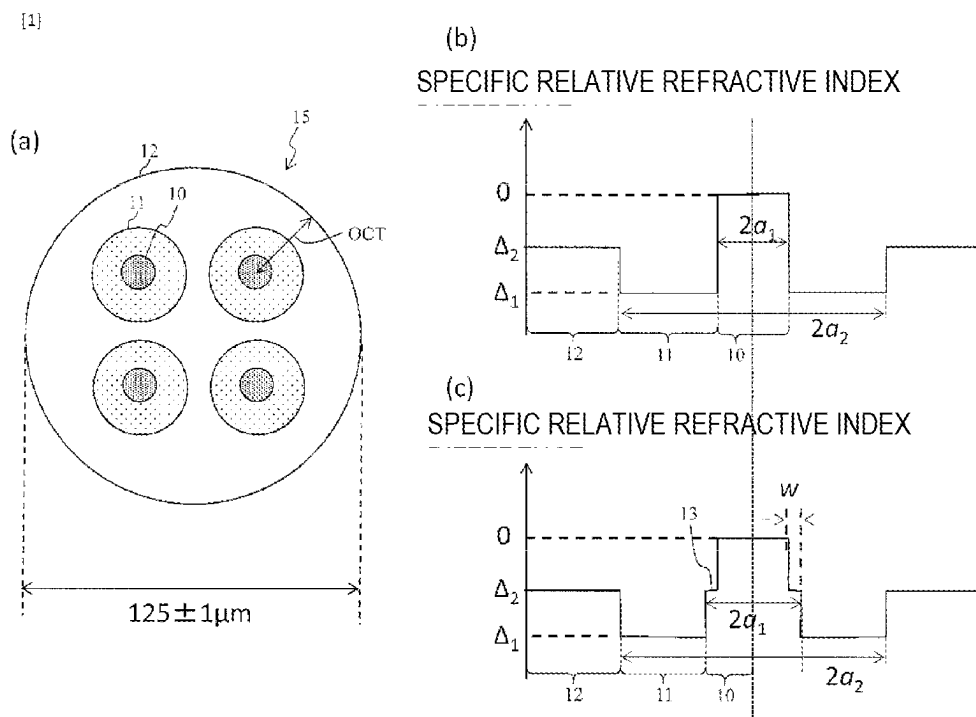
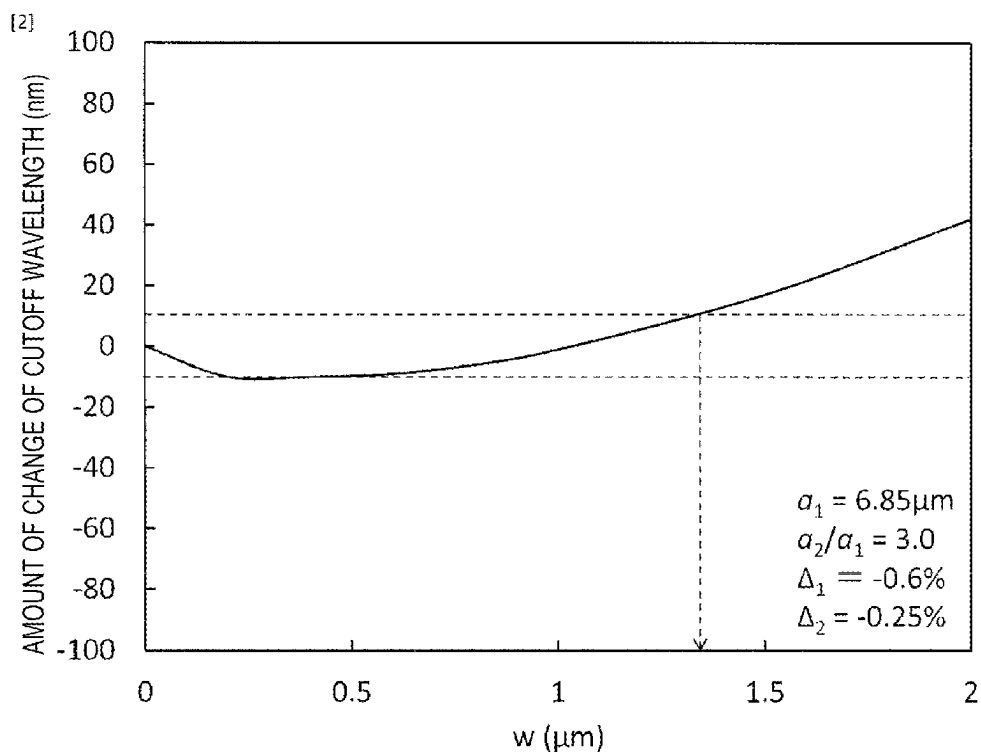
Fig. 1**Fig. 2**

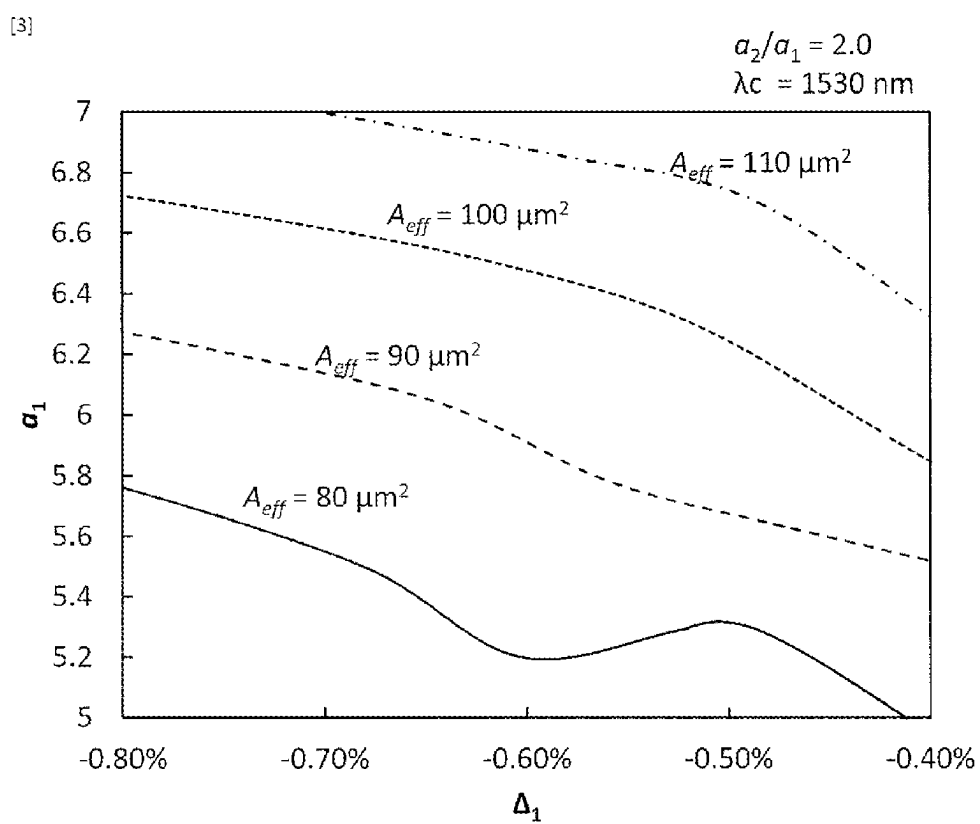
Fig. 3

Fig. 4

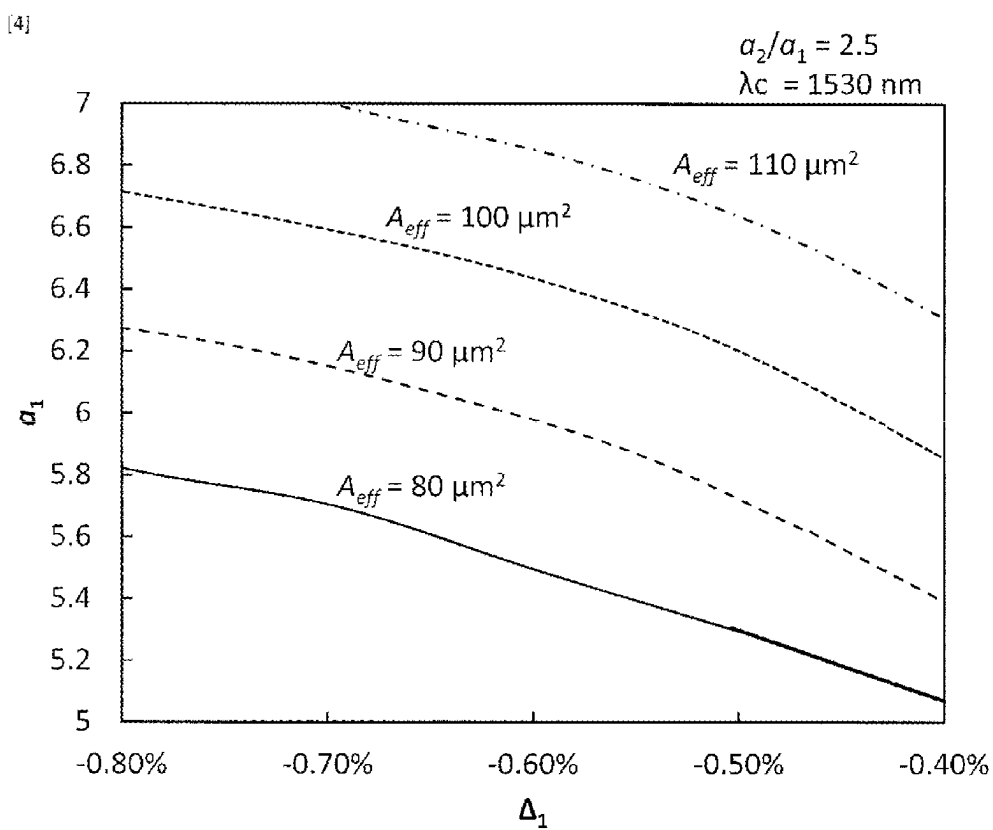


Fig. 5

[5]

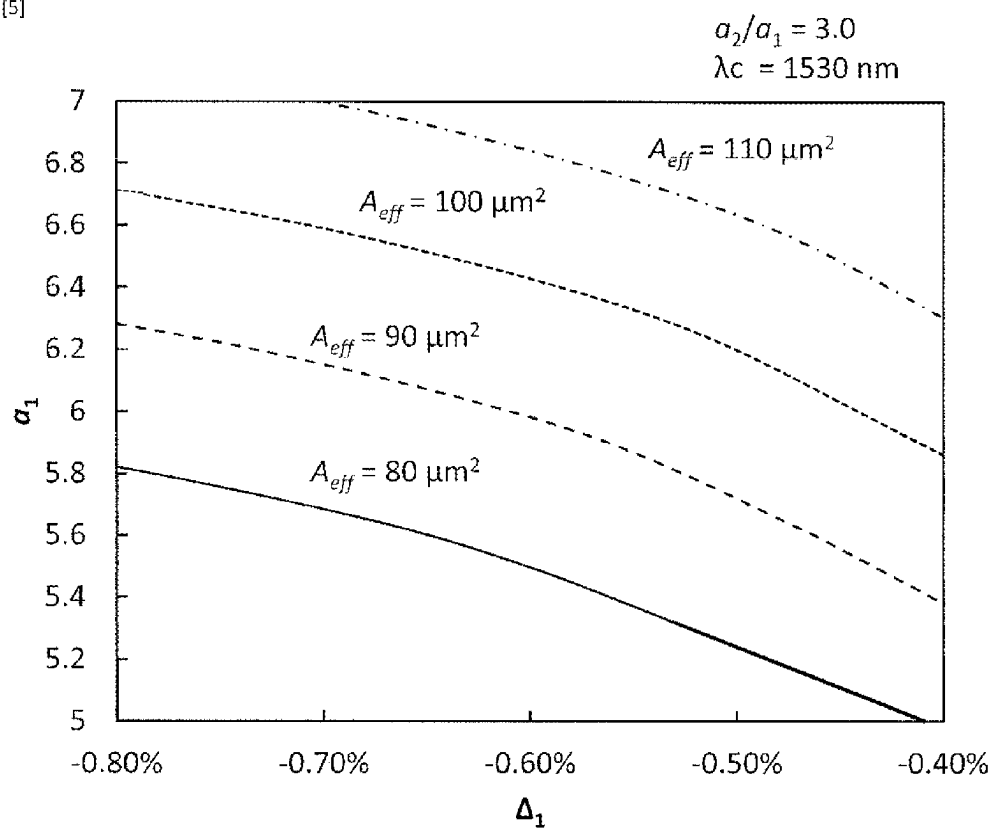


Fig. 6

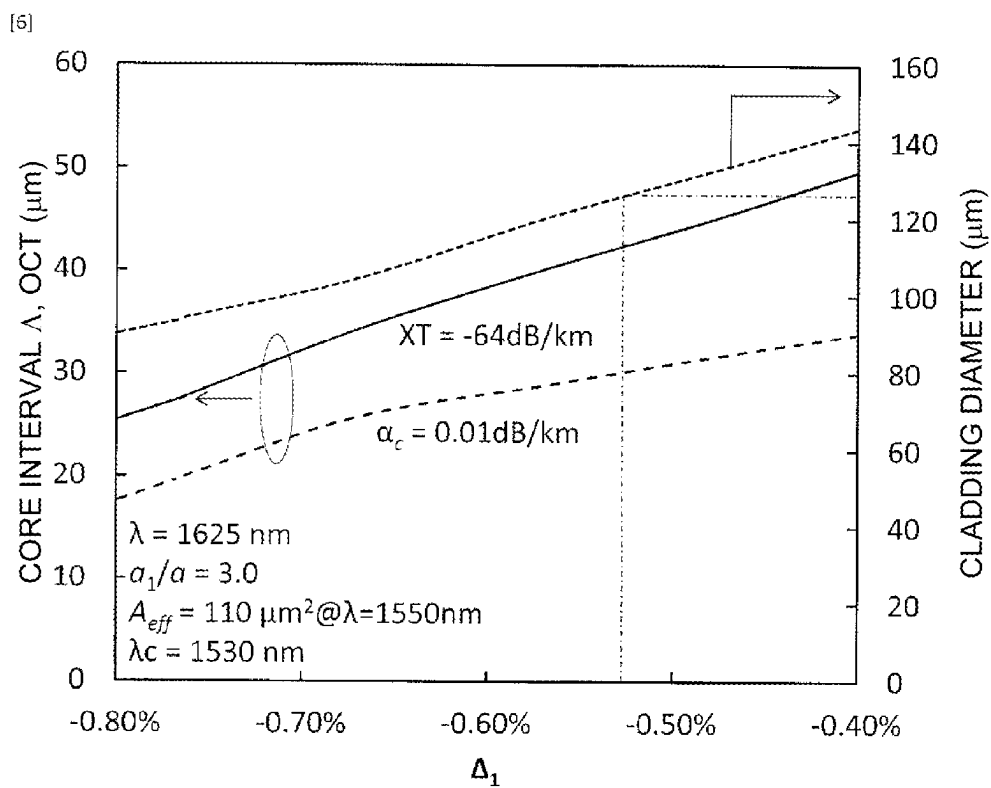


Fig. 7

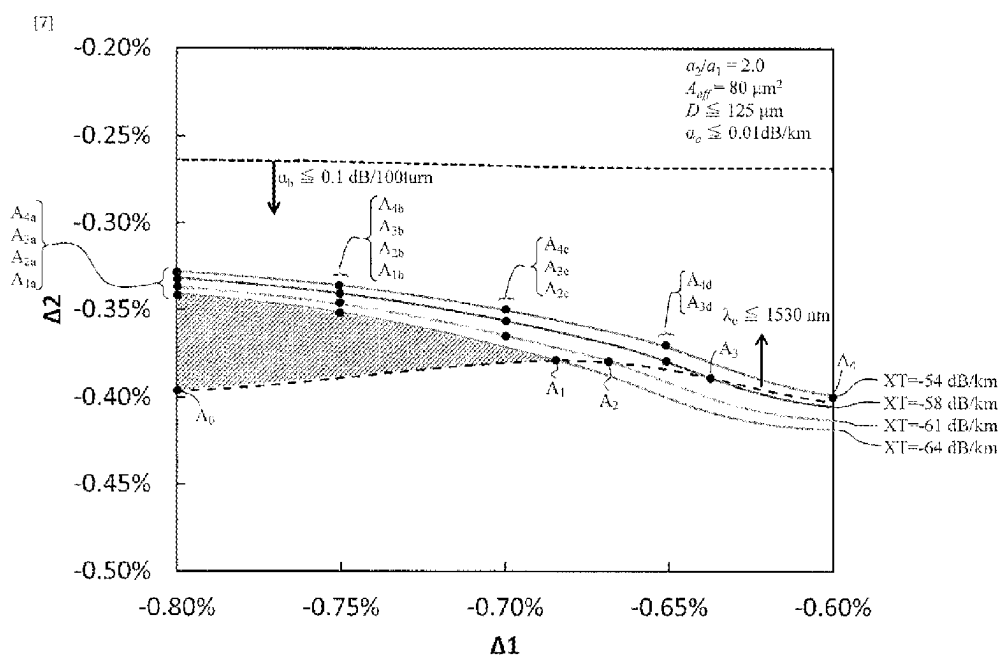


Fig. 8

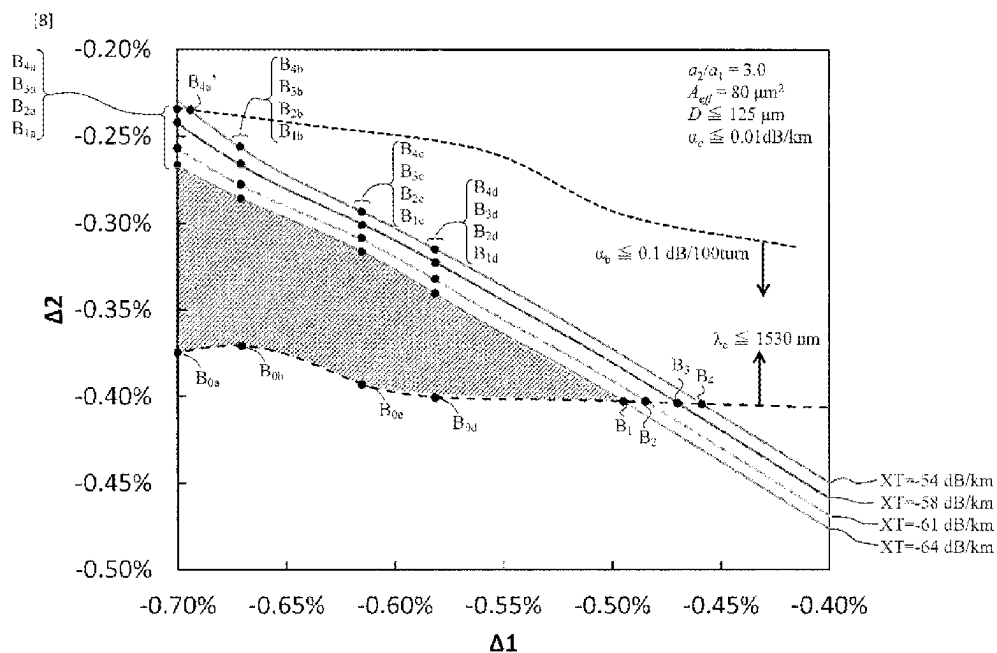


Fig. 9

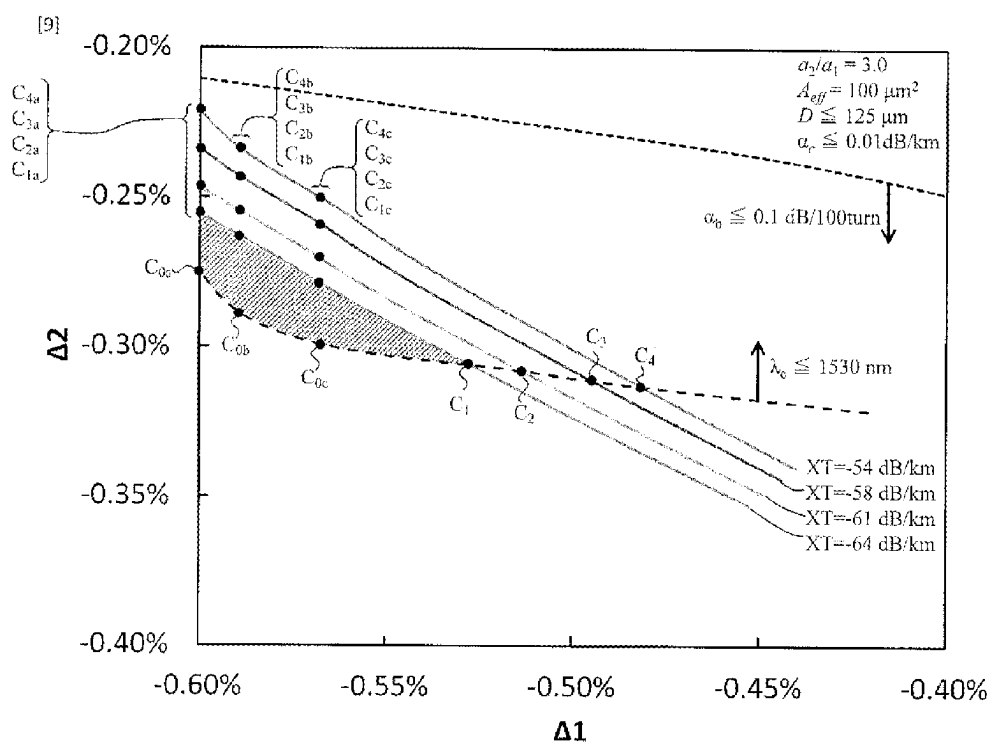


Fig. 10

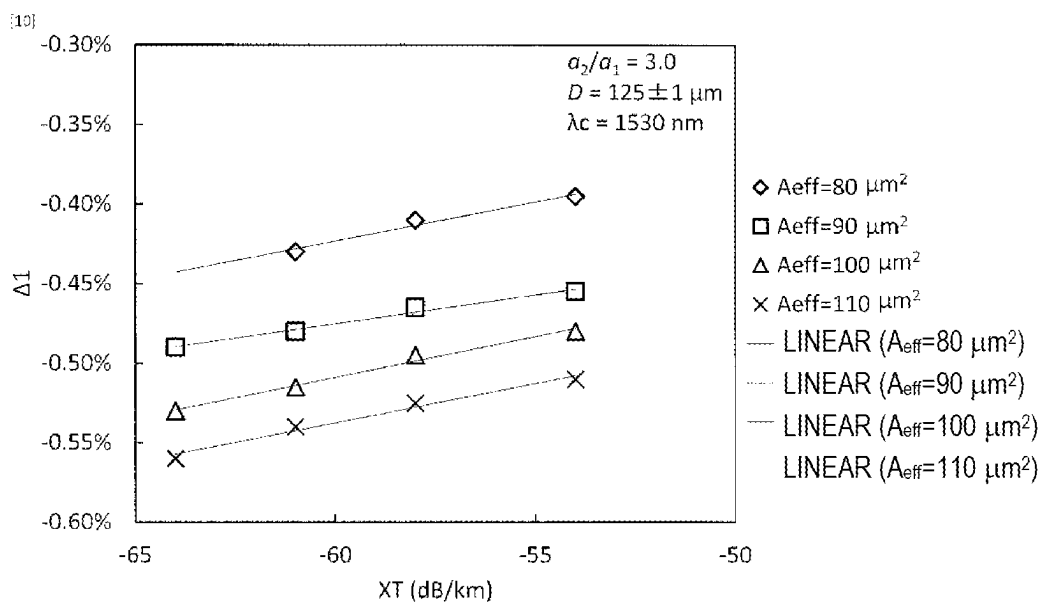


Fig. 11

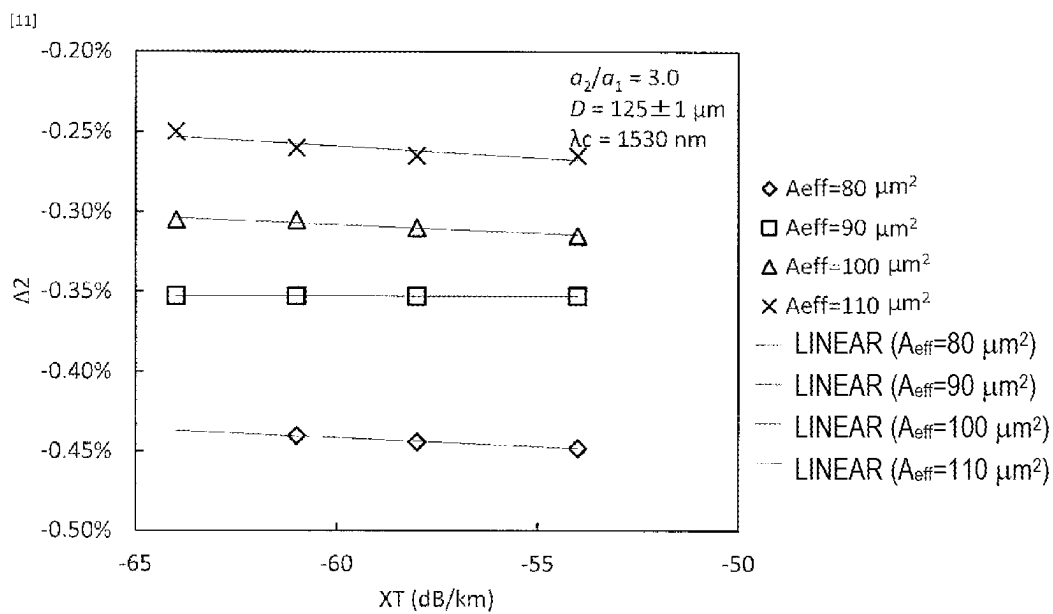


Fig. 12

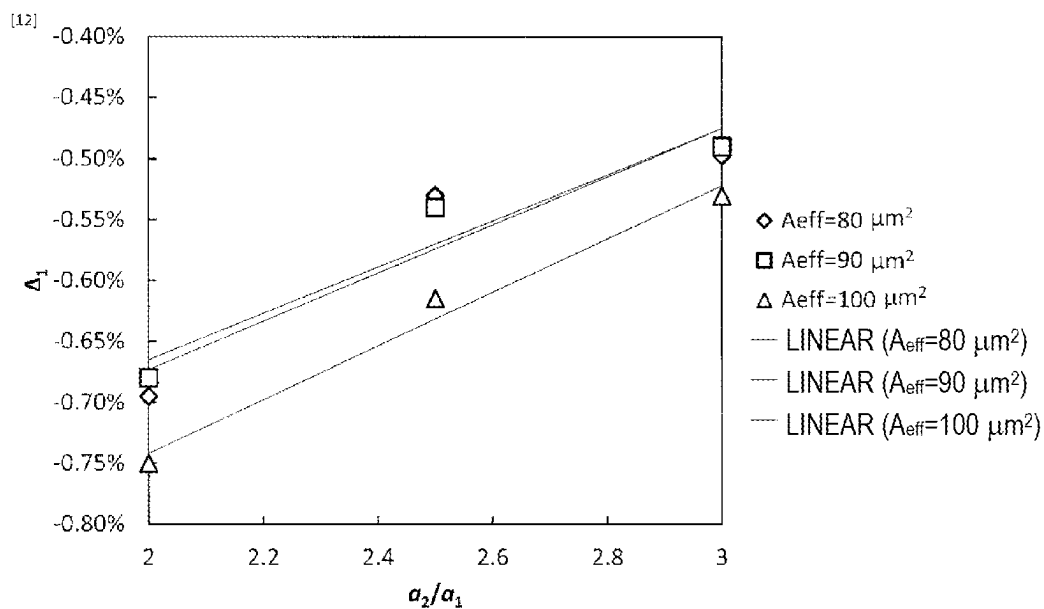


Fig. 13

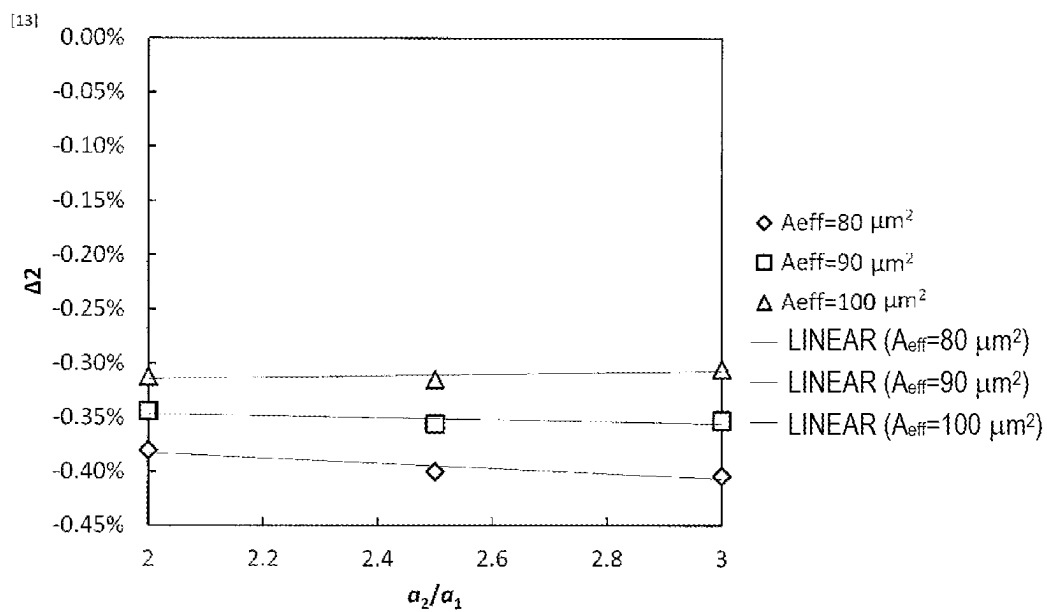


Fig. 14

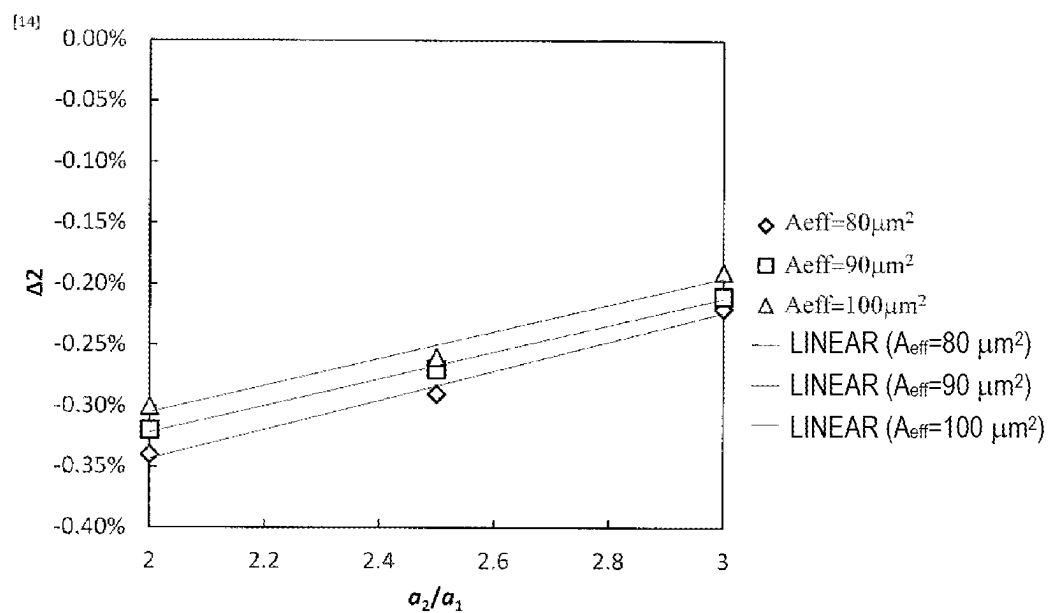


Fig. 15

[15]

		DESIGN EXAMPLE (1)	DESIGN EXAMPLE (2)
a_1		5.83 μm	6.57 μm
a_2/a_1		2.5	2.5
Δ_1		-0.68%	-0.61%
Δ_2		-0.38%	-0.31%
A_{eff}	$\lambda = 1550\text{nm}$	85 μm^2	102 μm^2
α_b	$\lambda = 1625\text{nm}$, $R=30\text{mm}$	<0.1dB/100turn	<0.1dB/100turn
λ_c		1530nm	1530nm

Fig. 16

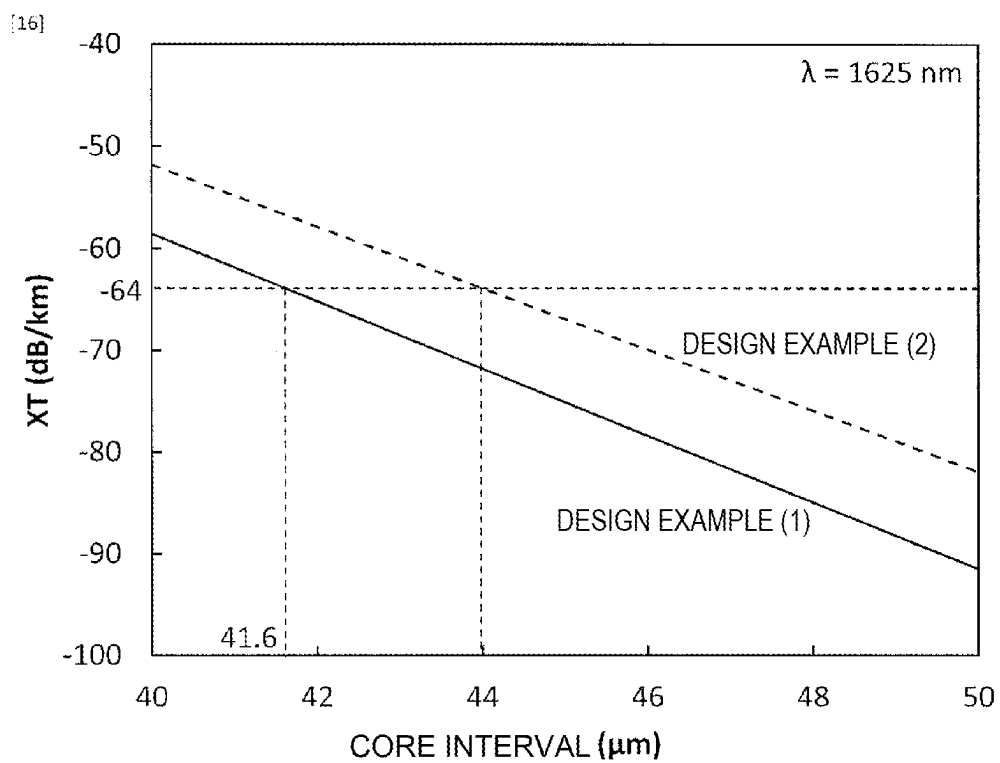


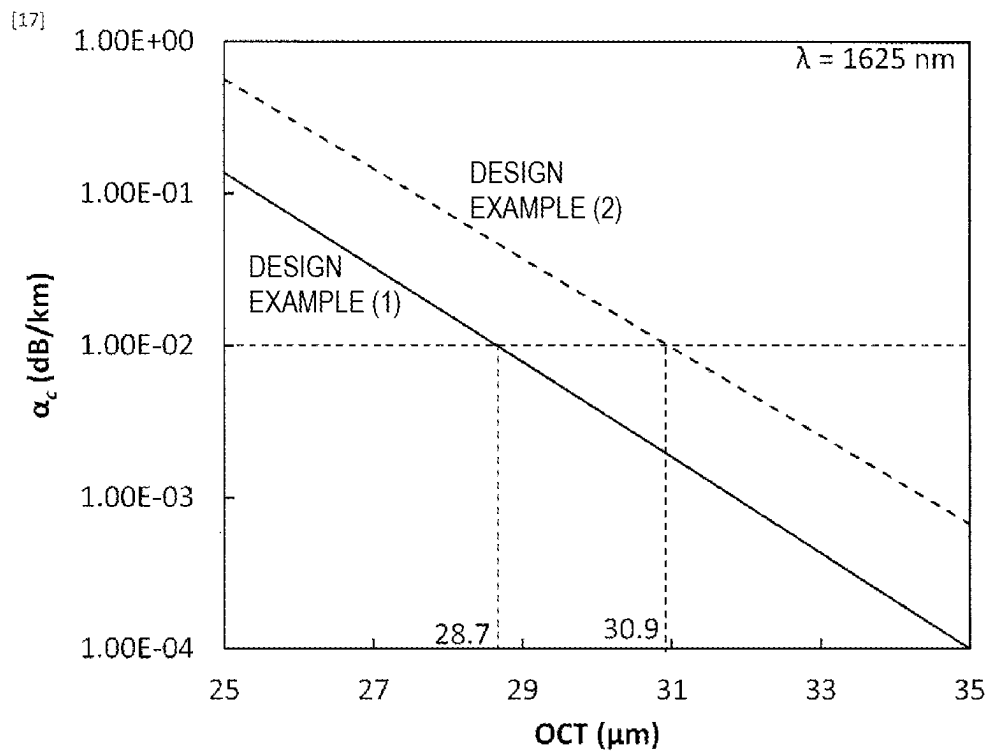
Fig. 17

Fig. 18

[18]

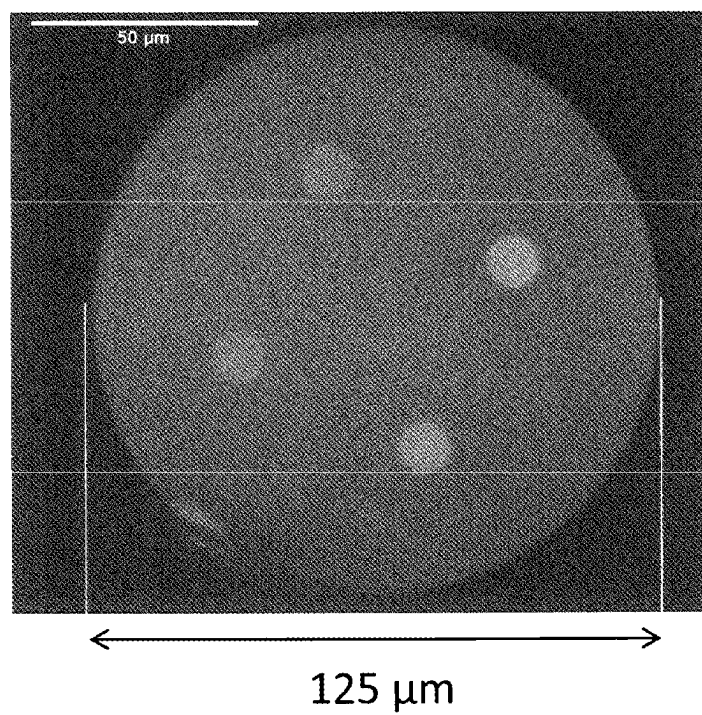
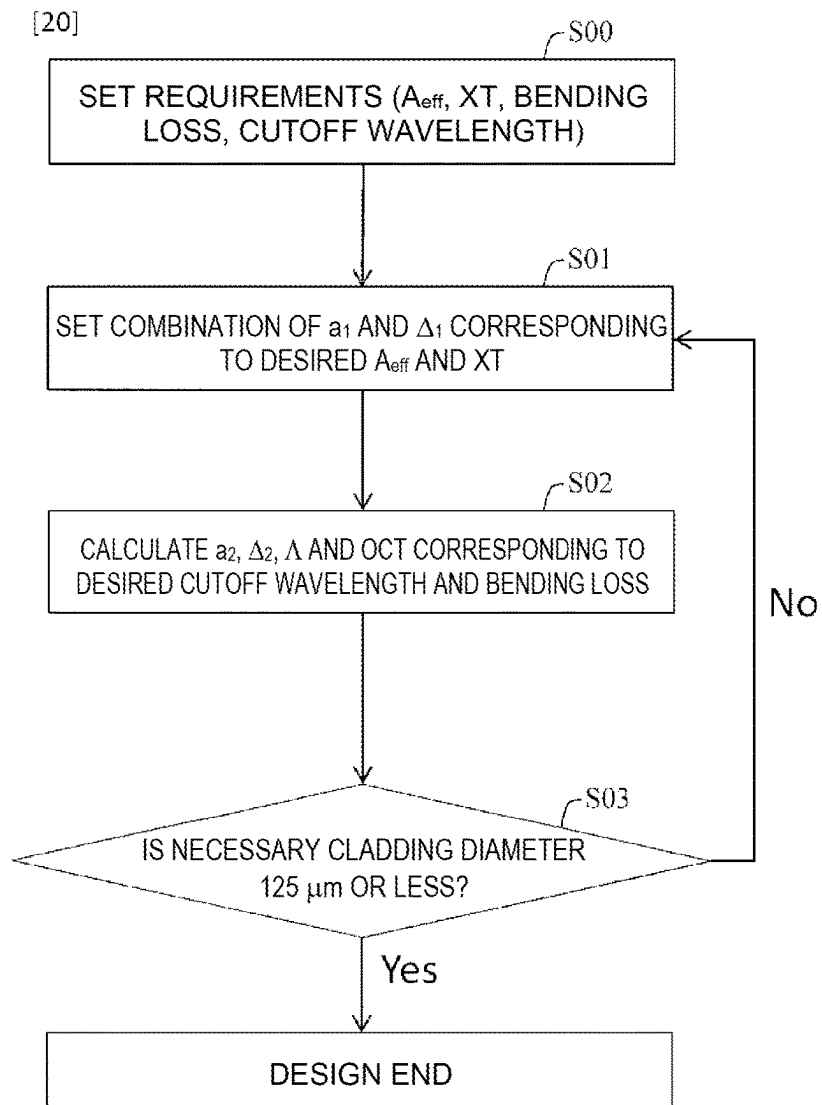
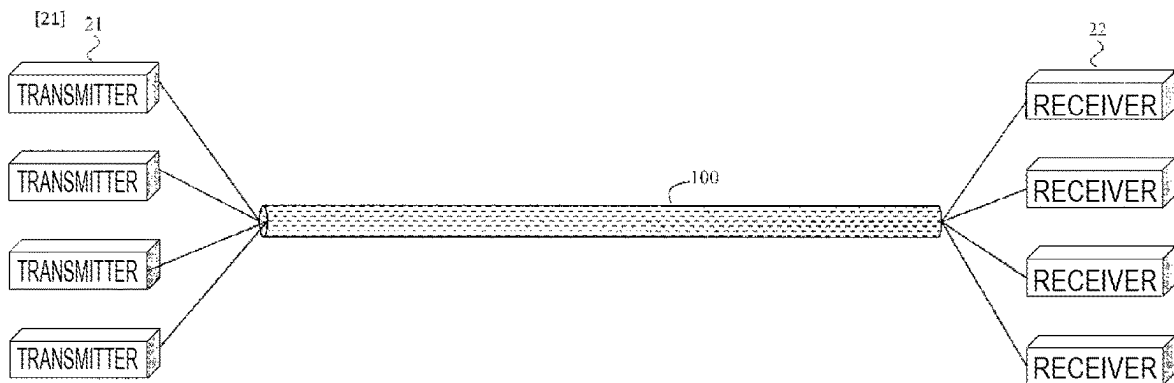


Fig. 19

[19]

	MEASUREMENT CONDITION	DESIGN EXAMPLE (1)
a_1		6 μ m
a_2/a_1		3
Δ_1		-0.6%
Δ_2		-0.4%
EFFECTIVE CROSS-SECTIONAL AREA A_{eff}	$\lambda = 1550\text{nm}$	83~89 μm^2
BENDING LOSS α_b	$\lambda = 1625\text{nm}$, R=30mm	<0.01 dB/100turn
CUTOFF WAVELENGTH λ_c		<1490 nm
TRANSMISSION LOSS	$\lambda = 1550\text{nm}$	<0.18 dB/km
	$\lambda = 1625\text{nm}$	<0.25 dB/km
XT	$\lambda = 1625\text{nm}$	<-66 dB/km

Fig. 20**Fig. 21**

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MULTICORE OPTICAL FIBER, METHOD OF DESIGNING MULTICORE OPTICAL FIBER, AND OPTICAL TRANSMISSION METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is continuation of U.S. application Ser. No. 17/294,892, filed May 18, 2021, which is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/JP2019/043826, having an International Filing Date of Nov. 8, 2019, which claims priority to Japanese Application Serial No. 2018-218153, filed on Nov. 21, 2018. The disclosures of the prior applications are considered part of the disclosure of this application, and are incorporated in their entirety into this application.

TECHNICAL FIELD

The present disclosure relates to a multicore optical fiber having a plurality of cores, a design method for the multicore optical fiber and an optical transmission method using the multicore optical fiber.

BACKGROUND ART

Multicore optical fibers having a plurality of core regions (MCF) are being actively studied in anticipation of drastic increase in transmission capacity using space division multiplexing technology. It is also known that power consumption can be reduced particularly in optical transmission paths using space division multiplexing, and MCF is expected to be used in long-distance optical communication systems including submarine communication systems; Non-Patent Literature 1 describes a case where a 10000 km class transmission experiment using MCF has been successfully conducted. Here, it is necessary to prevent deterioration of transmission characteristics in MCF due to inter-core crosstalk (XT). To reduce XT, not only a design of core parameters in MCF but also inter-core distances to be set appropriately are required, and Non-Patent Literature 2 reports on an MCF with an extremely low XT where inter-core XT is set to -30 dB or less in 10000 km.

Here, in order to set a sufficiently wide core interval to reduce XT, the MCF presented in many reports including Non-Patent Literatures 1 and 2 has a cladding diameter of 150 to 230 μm , larger than conventional optical fibers. However, since the length of an optical fiber manufactured from one optical fiber preform decreases in inverse proportion to the square of the cladding diameter, increasing the cladding diameter significantly degrades productivity of the optical fiber. Existing optical fiber parts or the like are designed in accordance with the conventional cladding diameter of 125 μm , and so peripheral parts need to be re-designed to utilize the MCF of the increased cladding diameter, requiring a lot of research and development for practical use.

Thus, MCF having a cladding diameter of 125 μm , which is equivalent to the conventional one, has been under development in recent years. Adopting a standard cladding diameter of 125 μm makes it possible to maintain mass productivity of optical fibers at the same level as the conventional level or higher, and utilize standard connection parts and existing peripheral articles such as optical cables. Furthermore, since each MCF core has optical characteristics equivalent to existing optical fiber, compatibility with exist-

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ing optical interfaces can be secured, it is possible to easily upgrade existing equipment to MCF.

Non-Patent Literatures 3 and 4 report on an MCF having XT of -30 dB or less in 100 km and including four cores with optical characteristics equivalent to existing single mode optical fiber (SMF). Non-Patent Literature 3 shows that four cores can be disposed when the same kind of core structure is used and Non-Patent Literature 5 shows that five cores can be disposed using a plurality of core structures. Non-Patent Literature 6 shows that MCF having characteristics equivalent to submarine low loss optical fibers is under study and two cores can be disposed at a cladding diameter of 125 μm .

CITATION LIST

Non-Patent Literature

- Non-Patent Literature 1: H. Takahashi et al., "First Demonstration of MC-EDFA-Repeatered SDM Transmission of 40x128-Gbit/s PDM-QPSK Signals per Core over 6,160-km 7-core MCF," ECOC2012, Th3C3, September 2012.
- Non-Patent Literature 2: T. Hayashi et al., "Design and fabrication of ultra-low crosstalk and low-loss multi-core fiber," Opt. Express, vol. 19, pp. 16576-16592, August 2011.
- Non-Patent Literature 3: T. Matsui et al., "Design of multicore fiber in 125 μm cladding diameter with full compliance to conventional SMF," ECOC2015, We.1.4.5, September 2015.
- Non-Patent Literature 4: T. Matsui et al., "118.5 Tbit/s Transmission over 316 km-Long Multi-Core Fiber with Standard Cladding Diameter" OECC2017, PDP2, August 2017.
- Non-Patent Literature 5: T. Gonda et al., "125 μm 5-core fibre with heterogeneous design suitable for migration from single-core system to multi-core system" ECOC2016, W2B1, September 2016.
- Non-Patent Literature 6: Tamura et al., "Aeff Expanded Non-Binding 2-Core Fiber Having Cladding Diameter of 125 μm " IEICE (The Institute of Electronics, Information and Communication Engineers) Society Conference, B-10-2, September 2016
- Non-Patent Literature 7: P. J. Winzer et al., "Penalties from In-Band Crosstalk for Advanced Optical Modulation Formats," ECOC2011, Tu5B7, September 2011.

SUMMARY OF THE INVENTION

Technical Problem

However, in consideration of MCF with four cores while maintaining an existing cladding diameter of 125 μm , there has been a problem of the limit being approximately several hundreds of kilometers due to the XT constraint as described in Non-Patent Literatures 3 to 5. As described in Non-Patent Literature 6, in order to realize an XT with which transmission of several thousands of kilometers can be supported, the number of cores is limited to two for a cladding diameter of 125 μm and it is difficult to dispose three or more cores.

In order to solve the above problems, an object of the present invention is to provide a multicore optical fiber with four cores having a standard cladding diameter of 125 $\pm$ 1 μm for an existing single mode optical fiber and covering several thousands of kilometers of transmission, a design

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method for the multicore optical fiber and an optical transmission method using the multicore optical fiber.

Means for Solving the Problem

In order to attain the above described object, a multicore optical fiber according to the present invention disposes two-stage claddings with different refractive indices disposed around each core, and designates a core radius a_1 , a radius a_2 of a first cladding region surrounding each core, specific refractive index Δ_1 with respect to the core of the first cladding region and a specific refractive index Δ_2 with respect to the core of a second cladding region including four cores and the first cladding regions as a predetermined range.

More specifically, the multicore optical fiber according to the present invention includes:

four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape;

a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , in which

an outer diameter of the second cladding region is $125 \pm 1 \mu\text{m}$,

an effective cross-sectional area at a wavelength of 1550 nm is $80 \mu\text{m}^2$ or more,

bending loss at a wavelength of 1625 nm and a bending radius of 30 μm is 0.1 dB/100 turns or less, and a cutoff wavelength is 1530 nm or less.

The presence of the first cladding region helps confine an electric field distribution to the core, making it possible to narrow a core interval with XT reduced and provide a multicore optical fiber having a cladding diameter of 125 μm with the four cores disposed, thus enabling long-distance transmission.

The multicore optical fiber according to the present invention may include a region having a width w and having the same refractive index as the refractive index of the second cladding region between the core and the first cladding region. In this case, the width w is preferably $0 \mu\text{m} < w \leq 1.3 \mu\text{m}$.

The parameters of the multicore optical fiber according to the present invention are as follows:

The relationship between the Δ_1 and the total value XT (dB/km) of inter-core crosstalk per km is characterized by satisfying Formula C1.

[Formula C1]

$$\Delta_1 \leq 4.93 \times 10^{-5} XT - 0.00127 \quad (\text{C1})$$

The relationship between the a_1 and the Δ_1 is characterized by satisfying Formula C2.

[Formula C2]

$$a_1 \geq -28041 \Delta_1^2 - 560.65 \Delta_1 + 3.1369 \quad (\text{C2})$$

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The relationship between the Δ_2 and the total value XT (dB/km) of inter-core crosstalk per km is characterized by satisfying Formula C3.

[Formula C3]

$$\Delta_2 \geq -1.14 \times 10^{-5} XT - 0.00509 \quad (\text{C3})$$

It is characterized that a ratio a_2/a_1 between the a_1 and the a_2 is 1.0 or more and 3.0 or less, the relationship between the Δ_1 , the a_2/a_1 , and the effective cross-sectional area A_{eff} at a wavelength of 1550 nm satisfies Formula C4, and the relationship between the Δ_2 , the a_2/a_1 , and the A_{eff} satisfies Formula C5.

[Formula C4]

$$\Delta_1 \leq (1.9 \times 10^{-5} A_{eff}^2 - 0.00033 A_{eff} + 0.0163) a_2/a_1 + (-8.0 \times 10^{-6} A_{eff}^2 + 0.00139 A_{eff} - 0.0705) \quad (\text{C4})$$

[Formula C5]

$$\Delta_2 \leq (5.0 \times 10^{-7} A_{eff}^2 + 6.5 \times 10^{-6} A_{eff} - 0.00108) a_2/a_1 + (-2.9 \times 10^{-7} A_{eff}^2 + 5.58 \times 10^{-5} A_{eff} - 0.00594)$$

and

$$\Delta_2 \leq (5.0 \times 10^{-7} A_{eff}^2 - 9.5 \times 10^{-5} A_{eff} + 0.0056) a_2/a_1 + (-1.6 \times 10^{-6} A_{eff}^2 + 0.00031 A_{eff} - 0.0208) \quad (\text{C5})$$

The Δ_1 and the Δ_2 are characterized by satisfying Formula C6.

[Formula C6]

$$-0.7\% \leq \Delta_1 \leq -0.49\%$$

$$-0.4\% \leq \Delta_2 \leq -0.27\% \quad (\text{C6})$$

The parameters of the multicore optical fiber according to the present invention can also be expressed as follows:

The present multicore optical fiber includes:

four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape;

a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , in which

$a_2/a_1 = 2.0$ and an effective cross-sectional area at a wavelength of 1550 nm is $80 \mu\text{m}^2$, and when the specific relative refractive index difference of the first cladding region is represented by an axis of abscissas and the specific relative refractive index difference of the second cladding region is represented by an axis of ordinates, it is characterized in that the Δ_1 and the Δ_2 are within a range enclosed by:

A0 (-0.800, -0.396)

A1a (-0.800, -0.340)

A1b (-0.750, -0.351)

A1 (-0.682, -0.378)

The present multicore optical fiber includes:

four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape;

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a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , in which

$a_2/a_1=3.0$ and an effective cross-sectional area at a wavelength of 1550 nm is $80 \mu\text{m}^2$, and

when the specific relative refractive index difference of the first cladding region is represented by an axis of abscissas and the specific relative refractive index difference of the second cladding region is represented by an axis of ordinates, it is characterized in that the Δ_1 and the Δ_2 are within a range enclosed by:

B0a (-0.700, -0.376)

B1a (-0.700, -0.266)

B1b (-0.671, -0.286)

B1c (-0.618, -0.313)

B1d (-0.586, -0.338)

B1 (-0.490, -0.396)

B0d (-0.586, -0.400)

B0c (-0.618, -0.391)

B0b (-0.671, -0.371)

The present multicore optical fiber includes:

four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape;

a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , in which

$a_2/a_1=3.0$ and an effective cross-sectional area at a wavelength of 1550 nm is $100 \mu\text{m}^2$, and when the specific relative refractive index difference of the first cladding region is represented by an axis of abscissas and the specific relative refractive index difference of the second cladding region is represented by an axis of ordinates, the Δ_1 and the Δ_2 are within a range enclosed by:

C0a (-0.600, -0.274)

C1a (-0.600, -0.255)

C1b (-0.590, -0.262)

C1c (-0.570, -0.277)

C1 (-0.528, -0.305)

C0c (-0.570, -0.297)

C0b (-0.590, -0.289)

The parameters of the aforementioned multicore optical fiber are designed as follows:

The present multicore optical fiber includes:

four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape;

a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the

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refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , the multicore optical fiber executing:

a first step of selecting a combination of the a_1 and the Δ_1 from requirements for effective cross-sectional area $A_{\text{eff}}(\text{m}^2)$ and a total value XT (dB/km) of inter-core crosstalk per km;

a second step of selecting the a_2 and the Δ_2 that satisfy requirements for a cutoff wavelength and bending loss in the combination selected in the first step and acquiring a core interval relationship between the Δ_1 with respect to the XT and a core interval Λ , and an OCT relationship between the Δ_1 and a shortest distance OCT from the outer periphery of the second cladding region to a center of the core with respect to requirements for an excessive loss α_c ; and

a third step of calculating an outer diameter ϕ of the second cladding region at the Δ_1 acquired in the first step using the core interval relationship and the OCT relationship acquired in the second step according to:

$$\phi=2\times(\Lambda\sqrt{2+OCT})$$

and determining whether the outer diameter ϕ becomes less than $125 \mu\text{m}$ or not.

Here, it is characterized in that when A_{eff} is $80 \mu\text{m}^2$ or more, the combination that satisfies Formula C1 and Formula C2 is selected in the first step. It is characterized in that when a_2/a_1 is 3.0 or less, the Δ_2 that satisfies Formula C3 is selected in the second step.

Furthermore, the optical transmission method according to the present invention is characterized in that

the multicore optical fiber of 1000 km or more is used as an optical transmission path,

signal light having a wavelength of 1530 nm or more is transmitted from four transmitters for each core of the multicore optical fiber, and

four receivers receive the signal light for each core of the multicore optical fiber.

The optical transmission method according to the present invention is characterized in that

a single mode optical fiber of a submarine communication system having a communication distance of 1000 km or more is replaced by the multicore optical fiber,

signal light having a wavelength of 1530 nm or more is transmitted from four transmitters for each core of the multicore optical fiber, and

four receivers receive the signal light for each core of the multicore optical fiber.

The multicore optical fiber according to the present invention includes four cores and realizes the second cladding having a diameter of $125 \mu\text{m}$ and XT of -54 dB/km or less. These are the characteristics unachieved by any one of the

multicore optical fibers according to Non-Patent Literatures 3 to 6. Thus, the present multicore optical fiber can replace

single mode optical fibers currently used in long-distance optical communication systems such as submarine communication systems, and since the multicore optical fiber is provided with four cores, the multicore optical fiber can drastically increase transmission capacity and reduce power consumption.

The present invention can provide a multicore optical fiber including four cores having a standard cladding diameter of $125 \pm 1 \mu\text{m}$ for existing single mode optical fibers and supporting transmission in several thousands of kilometers, a design method for the multicore optical fiber and an optical transmission method using the multicore optical fiber.

The present multicore optical fiber has effects of providing a high density, high capacity MCF suitable for ultra-long distance communication such as submarine systems, including four cores having a standard cladding diameter and capable of achieving optical characteristics having compatibility with existing optical fibers and realizing XT that enables transmission in several thousands of kilometers.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram illustrating an example of a structure of a multicore optical fiber according to the present invention.

FIG. 2 is a diagram illustrating influences of a width w of the multicore optical fiber according to the present invention.

FIG. 3 is a characteristic diagram expressing a relationship between an effective cross-sectional area and structural parameters of the multicore optical fiber according to the present invention.

FIG. 4 is a characteristic diagram expressing a relationship between an effective cross-sectional area and structural parameters of the multicore optical fiber according to the present invention.

FIG. 5 is a characteristic diagram expressing a relationship between an effective cross-sectional area and structural parameters of the multicore optical fiber according to the present invention.

FIG. 6 is a characteristic diagram expressing an example of a relationship between core arrangement, XT and excessive loss of the multicore optical fiber according to the present invention.

FIG. 7 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \pm 1 \mu\text{m}$ or less under a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 8 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \pm 1 \mu\text{m}$ or less under a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 9 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \pm 1 \mu\text{m}$ or less under a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 10 is a characteristic diagram expressing a relationship between XT and a range of necessary structural parameters of the multicore optical fiber according to the present invention.

FIG. 11 is a characteristic diagram expressing a relationship between XT and a range of necessary structural parameters of the multicore optical fiber according to the present invention.

FIG. 12 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \mu\text{m}$ or less under a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 13 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \mu\text{m}$ or less under a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 14 is a characteristic diagram expressing a relationship of structural parameters of the multicore optical fiber according to the present invention in which the cladding diameter is $125 \mu\text{m}$ or less under between a predetermined cutoff wavelength, bending loss and an effective cross-sectional area.

FIG. 15 is a table illustrating a design example of the multicore optical fiber according to the present invention.

FIG. 16 is a characteristic diagram illustrating a relationship between a core interval and XT in a design example of the multicore optical fiber according to the present invention.

FIG. 17 is a characteristic diagram illustrating a relationship between OCT and excessive loss in a design example of the multicore optical fiber according to the present invention.

FIG. 18 is a cross-sectional view of the multicore optical fiber according to the present invention.

FIG. 19 is a table describing evaluation results of the multicore optical fiber according to the present invention.

FIG. 20 is a flowchart describing a design method according to the present invention.

FIG. 21 is a diagram illustrating an optical transmission method according to the present invention.

DESCRIPTION OF EMBODIMENTS

Embodiments of the present invention will be described with reference to the accompanying drawings. The embodiments described hereinafter are examples of the present invention, and the present invention is not limited to the following embodiments. Note that components assigned the same reference numerals in the present Description and drawings denote the same components.

FIG. 1 is a diagram illustrating an example of a structure of an optical fiber 15 according to the present embodiment. FIG. 1(a) illustrates a cross-sectional structure, FIG. 1(b) and FIG. 1(c) illustrate a refractive index distribution of each core. The optical fiber 15 is a multicore optical fiber including:

- four cores 10 having a radius a_1 disposed in a longitudinal direction in square lattice shape;
- a first cladding region 11 disposed around the core 10 and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and
- a second cladding region 12 disposed on an outer periphery of the first cladding region 11 and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 .

In the optical fiber 15, an outer diameter of the second cladding region 12 is $125 \pm 1 \mu\text{m}$, an effective cross-sectional area at a wavelength of 1550 nm is $80 \mu\text{m}^2$ or more, bending loss at a wavelength of 1625 nm and a bending radius of 30 mm is 0.1 dB/100 turns or less, and a cutoff wavelength is 1530 nm or less.

The optical fiber 15 includes four cores in one optical fiber having a standard cladding diameter (outer diameter of the second cladding region 12) of $125 \pm 1 \mu\text{m}$.

As shown in FIG. 1(b), a refractive index distribution of each core includes the first cladding region 11 around the core 10 and having a refractive index lower than the refractive index of the core and includes the second cladding region 12 therearound and having a refractive index lower than the refractive index of the core and higher than the refractive index of the first cladding region. As shown in FIG. 1(c), a region 13 having a refractive index equivalent to the refractive index of the second cladding region 12 may also be included between the core 10 and the first cladding region 11.

Regarding the refractive index distribution in FIG. 1(c), FIG. 2 illustrates an amount of change in a cutoff wavelength with respect to a width w of the region 13. As shown in FIG. 2, even when the region 13 having a refractive index equivalent to the refractive index of the second cladding region 12 exists between the core 10 and the first cladding region 11, the amount of change in the cutoff wavelength is small. It can be confirmed from the diagram that when the width w is $1.3 \mu\text{m}$ or less, the amount of change in the cutoff wavelength is $\pm 10 \text{ nm}$, which is equivalent to a measurement error, and the refractive index distributions in FIG. 1(b) and FIG. 1(c) can be regarded as equivalent. Adopting the structure in FIG. 1(c) can reduce fluctuations in the core structure during manufacturing, and in the case of pure quartz, for example, the structure is more stable regarding the refractive index of the second cladding region, and it is possible to improve manufacturing deviation or yield, which is therefore preferable.

Note that although the specific relative refractive index difference is defined using the refractive index of the core as a reference here, the materials of the core and the cladding can be selected from among combinations of glass materials that can constitute the refractive index distribution shown in FIG. 1 such as a combination of pure quartz and fluorine added glass or a combination of GeO_2 added glass, pure quartz glass and fluorine added glass.

The "cutoff wavelength" of an optical fiber refers to a condition under which single mode propagation is possible, and light propagates in a single mode when the wavelength of the light is longer than the cutoff wavelength or light propagates in a multimode when the wavelength of the light is shorter than the cutoff wavelength.

FIG. 3 to FIG. 5 are diagrams illustrating a design range of core parameters to obtain a predetermined effective cross-sectional area A_{eff} of the optical fiber 15 at a wavelength of 1550 nm. The axis of abscissas represents a specific relative refractive index difference Δ_1 with respect to the core of the first cladding region 11 and the axis of ordinates represents a radius a_1 of the core 10. FIG. 3 to FIG. 5 illustrate cases where the ratio a_2/a_1 of the core radius a_1 to the radius a_2 of the first cladding region 11 region is changed from 2.0 to 2.5 to 3.0 respectively. The specific relative refractive index difference Δ of the cladding with respect to the core is set so that the cutoff wavelength becomes 1530 nm.

Comparing FIG. 3 to FIG. 5, the relationship between a_1 and Δ_1 to obtain constant A_{eff} is substantially the same. It is considered that this is because the electric field distribution is confined in a region surrounded by the first cladding region 11, and so dependency on the parameters a_2 and Δ_2 (specific relative refractive index difference of the second cladding region 12 with respect to the core) relating to the second cladding region 12 is small.

Here, a conventional SMF has an A_{eff} of approximately $80 \mu\text{m}^2$ at a wavelength of 1550 nm and the A_{eff} is preferably $80 \mu\text{m}^2$ or more to secure connectivity and low non-linearity equal to or better than conventional. From FIG. 3 to FIG. 5, in order to set A_{eff} to $80 \mu\text{m}^2$,

[Formula 1]

$$a_1 \approx -28041\Delta_1^2 - 560.65\Delta_1 + 3.1369 \quad (1)$$

needs to be satisfied according to an approximate curve shown by the solid line.

It is known that the core interval needs to be increased by a certain amount in MCF to reduce XT. Deterioration of transmission characteristics caused by XT depends on a transmission scheme and Non-Patent Literature 7 shows that crosstalk should be reduced to -18 dB, -24 dB and -32 dB or less for QPSK, 16QAM and 64QAM respectively. In the case of a multicore optical fiber, inter-core crosstalk is a sum total of crosstalk components from other cores to a given core and crosstalk at a given distance with respect to crosstalk (XT, unit dB/km) per unit distance and a distance L (km) is given by $\text{XT} + \log(L)$.

Therefore, in order to perform transmission in 1000 to 10000 km using, for example, a 16QAM signal format, crosstalk of a multicore optical fiber needs to be -54 to -64 dB/km or less. Reducing crosstalk requires the core interval to be increased by a certain amount, but it is known that when the cladding diameter is constant, a thickness (OCT) between a core and a cladding end decreases as the core interval increases, an excessive loss α_c is generated due to light wave leakage.

FIG. 6 is a diagram illustrating a design example of the optical fiber 15 to reduce inter-core crosstalk (XT) and excessive loss α_c . The axis of abscissas represents a specific relative refractive index difference Δ_1 of the first cladding region 11, a first axis of ordinates represents a core interval Λ or OCT, and a second axis of ordinates represents an outer diameter of the second cladding region 12. Since XT and the excessive loss increase as the wavelength increases, the wavelength is assumed to be 1625 nm in consideration of utilization of an entire C+L bandwidth. The solid line shows a core interval Λ (μm) where XT becomes -64 dB/km and the broken line shows OCT where the excessive loss α_c becomes 0.01 dB/km. That is, XT becomes less than -64 dB/km in a region above the solid line and α_c becomes less than 0.01 dB/km in a region above the broken line.

The dotted line shows a "necessary cladding diameter" for XT to satisfy -64 dB/km and for α_c to satisfy 0.01 dB/km, and in the case of a 4-core arrangement, the necessary cladding diameter = $2 \times (\Lambda/\sqrt{2} + \text{OCT})$.

Here $a_2/a_1 = 3.0$ and a_1 and Δ_2 are set so that A_{eff} becomes $110 \mu\text{m}^2$ and the cutoff wavelength λ_c becomes 1530 nm.

It is seen from FIG. 6 that light wave confinement becomes weak as Δ_1 increases, and the core interval Λ and OCT necessary to obtain a desired XT and α_c increase. In the case of FIG. 6, Δ_1 needs to be -0.52% in order to set the cladding diameter to $125 \pm 1 \mu\text{m}$ (see single-dot dashed line). The necessary core interval Λ and OCT at this time are 42.5 μm and 30 μm respectively. Note that the core interval and

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the OCT shown in FIG. 6 are minimum values that satisfy the predetermined XT and the excessive loss, and setting larger values will cause the XT and the excessive loss to have smaller values. That is, if Δ_1 is set to less than -0.52% in order to set the cladding diameter to $125 \pm 1 \mu\text{m}$, the necessary cladding diameter becomes $125 \mu\text{m}$ or less, providing room for the design of Λ and OCT.

FIG. 7 to FIG. 9 are diagrams illustrating examples of structural parameters of the optical fiber 15 having a cutoff wavelength of 1530 nm or less, bending loss of $0.1 \text{ dB}/100$ turns or less at a wavelength of 1625 nm and a bending radius of $30 \mu\text{m}$ and a necessary cladding diameter of $125 \pm 1 \mu\text{m}$. In all the drawings, the aforementioned conditions (XT is $-64 \text{ dB}/\text{km}$ or less and α_c is $0.01 \text{ dB}/\text{km}$ or less) can be satisfied simultaneously in the shaded regions.

FIG. 7 shows a case where A_{eff} is $80 \mu\text{m}^2$ at a wavelength of 1550 nm , and $a_2/a_1=2.0$,

FIG. 8 shows a case where A_{eff} is $80 \mu\text{m}^2$ at a wavelength of 1550 nm , and $a_2/a_1=3.0$, and

FIG. 9 shows a case where A_{eff} is $110 \mu\text{m}^2$ at a wavelength of 1550 nm , and $a_2/a_1=3.0$.

In FIG. 7 to FIG. 9, the broken line shows a condition under which the cutoff wavelength becomes 1530 nm (the cutoff wavelength is 1530 nm or less in a region above the broken line) and the dotted line shows a condition under which the bending loss α_b becomes $0.1 \text{ dB}/100$ turns or less in a region below the dotted line).

In FIG. 7 to FIG. 9, the solid line shows a condition of XT at a wavelength of 1625 nm and XT becomes $-54 \text{ dB}/\text{km}$ or less, $-58 \text{ dB}/\text{km}$ or less, $-61 \text{ dB}/\text{km}$ or less and $-64 \text{ dB}/\text{km}$ or less respectively in a region below the solid line.

The region that satisfies all the conditions is a region surrounded by the solid line, the broken line and the dotted line, and when, for example, XT is set to $-64 \text{ dB}/\text{km}$ or less, the region corresponds to a shaded region in the diagram. From FIG. 7 to FIG. 9, an upper limit of Δ_1 and a lower limit of Δ_2 are determined by requirements for XT and cutoff wavelength, and the lower limit of Δ_2 has a relatively smaller change than Δ_1 . Therefore, the upper limit of Δ_1 and the lower limit of Δ_2 can be given by an intersection of the solid line and the broken line. It is also seen from FIG. 7 to FIG. 9 that the designable region becomes alleviated as the a_2/a_1 is larger and the required A_{eff} is smaller.

Therefore, when A_{eff} is $80 \mu\text{m}^2$ or more, the cutoff wavelength is 1530 nm or less and XT is $-64 \text{ dB}/\text{km}$ or less, FIG. 8 includes the largest region, and in this case,

[Formula 2]

$$-0.7\% \leq \Delta_1 \leq -0.49\%$$

$$-0.4\% \leq \Delta_2 \leq -0.27\%$$

(2)

In FIG. 7, if X coordinate is assumed to be Δ_1 and Y coordinate is assumed to be Δ_2 , the shaded region, that is, the region that satisfies:

$a_2/a_1=2.0$,
 $A_{\text{eff}}=80 \mu\text{m}^2$ at a wavelength of 1550 nm ,
cladding diameter $D \leq 125 \mu\text{m}$,
excessive loss $\alpha_c \leq 0.01 \text{ dB}/\text{km}$,
cutoff wavelength $\lambda_c \leq 1530 \text{ nm}$,
bending loss $\alpha_b \leq 0.1 \text{ dB}/100$ turns, and
 $XT \leq -64 \text{ dB}/\text{km}$ is a range surrounded by:
A0 ($-0.800, -0.396$)
A1a ($-0.800, -0.340$)
A1b ($-0.750, -0.351$)
A1 ($-0.682, -0.378$).

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Note that if the region satisfies $XT \leq -61 \text{ dB}/\text{km}$, it is a range surrounded by:

A0 ($-0.800, -0.396$)
A2a ($-0.800, -0.335$)
A2b ($-0.750, -0.346$)
A2c ($-0.700, -0.363$)
A2 ($-0.668, -0.380$).

If the region satisfies $XT \leq -58 \text{ dB}/\text{km}$, it is a range surrounded by:

A0 ($-0.800, -0.396$)
A3a ($-0.800, -0.331$)
A3b ($-0.750, -0.339$)
A3c ($-0.700, -0.355$)
A3d ($-0.650, -0.380$)
A3 ($-0.639, -0.387$).

If the region satisfies $XT \leq -54 \text{ dB}/\text{km}$, it is a range surrounded by:

A0 ($-0.800, -0.396$)
A4a ($-0.800, -0.328$)
A4b ($-0.750, -0.323$)
A4c ($-0.700, -0.349$)
A4d ($-0.650, -0.370$)
A4 ($-0.600, -0.400$).

In FIG. 8, if X coordinate is assumed to be Δ_1 and Y coordinate is assumed to be Δ_2 , the shaded region, that is, the region that satisfies:

$a_2/a_1=3.0$,
 $A_{\text{eff}}=80 \mu\text{m}^2$ at a wavelength of 1550 nm ,
cladding diameter $D \leq 125 \mu\text{m}$,
excessive loss $\alpha_c \leq 0.01 \text{ dB}/\text{km}$,
cutoff wavelength $\lambda_c \leq 1530 \text{ nm}$,
bending loss $\alpha_b \leq 0.1 \text{ dB}/100$ turns, and
 $XT \leq -64 \text{ dB}/\text{km}$ is a range surrounded by:

B0a ($-0.700, -0.376$)
B1a ($-0.700, -0.266$)
B1b ($-0.671, -0.286$)
B1c ($-0.618, -0.313$)
B1d ($-0.586, -0.338$)
B1 ($-0.490, -0.396$)
B0d ($-0.586, -0.400$)
B0c ($-0.618, -0.391$)
B0b ($-0.671, -0.371$).

Note that if the region satisfies $XT \leq -61 \text{ dB}/\text{km}$, it is a range surrounded by:

B0a ($-0.700, -0.376$)
B2a ($-0.700, -0.256$)
B2b ($-0.671, -0.278$)
B2c ($-0.618, -0.306$)
B2d ($-0.586, -0.329$)
B2 ($-0.484, -0.396$)
B0d ($-0.586, -0.400$)
B0c ($-0.618, -0.391$)
B0b ($-0.671, -0.371$).

If the region that satisfied $XT \leq -58 \text{ dB}/\text{km}$, it is a range surrounded by:

B0a ($-0.700, -0.376$)
B3a ($-0.700, -0.242$)
B3b ($-0.671, -0.267$)
B3c ($-0.618, -0.299$)
B3d ($-0.586, -0.319$)
B3 ($-0.470, -0.396$)
B0d ($-0.586, -0.400$)
B0c ($-0.618, -0.391$)
B0b ($-0.671, -0.371$).

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If the region that satisfies $XT \leq -54$ dB/km, it is a range surrounded by:

- B0a (-0.700, -0.376)
- B4a (-0.700, -0.234)
- B4b (-0.671, -0.256)
- B4c (-0.618, -0.278)
- B4d (-0.586, -0.312)
- B4 (-0.458, -0.396)
- B0d (-0.586, -0.400)
- B0c (-0.618, -0.391)
- B0b (-0.671, -0.371).

In FIG. 9, if X coordinate is assumed to be Δ_1 and Y coordinate is assumed to be Δ_2 , the shaded region, that is, the region that satisfies:

- $a_2/a_1 = 3.0$,
- $A_{eff} = 110 \mu m^2$ at a wavelength of 1550 nm
- cladding diameter $D \leq 125 \mu m$,
- excessive loss $\alpha_c \leq 0.01$ dB/km,
- cutoff wavelength $\lambda_c \leq 1530$ nm,
- bending loss $\alpha_b \leq 0.1$ dB/100 turns, and
- $XT \leq -64$ dB/km

is a region surrounded by:

- C0a (-0.600, -0.274)
- C1a (-0.600, -0.255)
- C1b (-0.590, -0.262)
- C1c (-0.570, -0.277)
- C1 (-0.528, -0.305)
- C0c (-0.570, -0.297)
- C0b (-0.590, -0.289).

Note that If the region that satisfies $XT \leq -61$ dB/km, it is a region surrounded by:

- C0a (-0.600, -0.274)
- C2a (-0.600, -0.247)
- C2b (-0.590, -0.254)
- C2c (-0.570, -0.269)
- C2 (-0.513, -0.308)
- C0c (-0.570, -0.297)
- C0b (-0.590, -0.289).

If the region that satisfies $XT \leq -58$ dB/km, it is a region surrounded by:

- C0a (-0.600, -0.274)
- C3a (-0.600, -0.234)
- C3b (-0.590, -0.242)
- C3c (-0.570, -0.257)
- C3 (-0.495, -0.311)
- C0c (-0.570, -0.297)
- C0b (-0.590, -0.289).

If the region that satisfies $XT \leq -54$ dB/km, it is a region surrounded by:

- C0a (-0.600, -0.274)
- C4a (-0.600, -0.220)
- C4b (-0.590, -0.233)
- C4c (-0.570, -0.249)
- C4 (-0.479, -0.313)
- C0c (-0.570, -0.297)
- C0b (-0.590, -0.289).

FIG. 10 and FIG. 11 are diagrams illustrating dependency on XT of the upper limit of Δ_1 (FIG. 10) and the lower limit of Δ_2 (FIG. 11) given by intersections of the cutoff wavelength λ_c (broken line) and the condition of XT (solid line) on the graphs created as shown in FIG. 7 to FIG. 9. Here, a_2/a_1 is assumed to be 3.0. It is seen from FIG. 10 and FIG. 11 that the upper limit of Δ_1 and the lower limit of Δ_2 linearly

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change with respect to the requirements for XT. If A_{eff} is $80 \mu m^2$ or more, it is seen from FIG. 10 and FIG. 11 that:

[Formula 3]

$$\Delta_1 \leq 4.93 \times 10^{-5} XT - 0.00127$$

$$\Delta_2 \geq -1.14 \times 10^{-5} XT - 0.00509$$

Here, comparing FIG. 7 and FIG. 8, the upper limit of Δ_1 decreases and the lower limit of Δ_2 increases as a_2/a_1 decreases, and so when a_2/a_1 is smaller than 3.0, a_2/a_1 is included within ranges of the upper limit of Δ_1 and the lower limit of Δ_2 shown in Formula 3.

FIG. 12 to FIG. 14 are diagrams illustrating dependency on a_2/a_1 of the upper limit of Δ_1 (FIG. 12), the lower limit of Δ_2 (FIG. 13) and the upper limit of Δ_2 (FIG. 14) given by intersections of the cutoff wavelength λ_c (broken line) and the condition of XT (solid line) on the graphs created as shown in FIG. 7 to FIG. 9. Straight lines in the respective drawings are the results of linear approximation.

Here, the requirements for XT are assumed to be -64 dB/km. It is seen from FIG. 12 to FIG. 14 that the upper limit of Δ_1 , the lower limit of Δ_2 (Δ_{2_min}) and the upper limit of Δ_2 (Δ_{2_max}) can be approximated relatively better with respect to a_2/a_1 through linear approximation. It is seen from FIG. 12 to FIG. 14 that:

With respect to:

$$A_{eff} = 80 \mu m^2,$$

[Formula 4]

$$\Delta_1 = 0.0020 a_2/a_1 - 0.01069$$

$$\Delta_{2_min} = -0.00024 a_2/a_1 - 0.00335$$

$$\Delta_{2_max} = 0.0012 a_2/a_1 - 0.005833$$

With respect to:

$$A_{eff} = 90 \mu m^2,$$

[Formula 5]

$$\Delta_1 = 0.0019 a_2/a_1 - 0.01045$$

$$\Delta_{2_min} = -9 \times 10^{-5} a_2/a_1 - 0.00329$$

$$\Delta_{2_max} = 0.0011 a_2/a_1 - 0.005417$$

With respect to:

$$A_{eff} = 100 \mu m^2,$$

[Formula 6]

$$\Delta_1 = 0.0022 a_2/a_1 - 0.01182$$

$$\Delta_{2_min} = -7 \times 10^{-5} a_2/a_1 - 0.00328$$

$$\Delta_{2_max} = 0.0011 a_2/a_1 - 0.00525.$$

When approximate curves of coefficients of Formula 4 to Formula 6 are taken,

[Formula 7]

$$\Delta_1 \leq (1.9 \times 10^{-6} A_{eff}^2 - 0.00033 A_{eff} + 0.0163) a_2/a_1 + (-8.0 \times 10^{-6} A_{eff}^2 + 0.00139 A_{eff} - 0.0705)$$

$$\Delta_2 \geq (5.0 \times 10^{-8} A_{eff}^2 + 6.5 \times 10^{-6} A_{eff} - 0.00108) a_2/a_1 + (-2.9 \times 10^{-7} A_{eff}^2 + 5.58 \times 10^{-5} A_{eff} - 0.00594)$$

$$\Delta_2 \leq (5.0 \times 10^{-7} A_{eff}^2 - 9.5 \times 10^{-5} A_{eff} + 0.0056) a_2/a_1 + (-1.6 \times 10^{-6} A_{eff}^2 + 0.00031 A_{eff} - 0.0208)$$

Within the parameter range that satisfies Formula 7, it is possible to realize an optical fiber having a standard clad-

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ding diameter of 125 μm , with four cores and having characteristics equivalent to existing optical fibers.

FIG. 15 shows structural parameters and optical characteristics designed within the design range of Formula 7. The 4-core optical fibers in design examples (1) and (2) have A_{eff} of 85 μm^2 and 102 μm^2 , acquire single mode operation in a C band and an L band, and have bending loss equivalent to or less than conventional SMF.

FIG. 16 and FIG. 17 show XT characteristics and excessive loss corresponding to the design example shown in FIG. 15. The wavelength is 1625 nm.

FIG. 16 is a diagram illustrating a relationship between a core interval and XT. In order to obtain XT of -64 dB/km or less, core intervals of 41.6 μm or less and 44 μm or less are necessary with respect to design example (1) and design example (2) respectively.

FIG. 17 illustrates a relationship between OCT and excessive loss α_c . In order to obtain excessive loss of 0.01 dB/km or less, OCTs of 28.7 μm or less and 30.9 μm or less are necessary with respect to design example (1) and design example (2) respectively.

It is seen from these results that in the 4-core optical fibers in design example (1) and design example (2), the minimum necessary cladding diameters are 116.2 μm and 124.0 μm respectively. It is seen in both design examples that four cores having a standard cladding diameter of 125 μm can be disposed.

EXAMPLES

FIG. 18 is a cross-sectional photo of the multicore optical fiber according to the present example. FIG. 19 shows the measurement results.

It is seen from the cross-sectional photo in FIG. 18 that the multicore optical fiber has a cladding diameter of 125 μm and that four cores are disposed.

It can also be confirmed from FIG. 19 that in a refractive index distribution of each core, a core radius is 6 μm , a_2/a_1 is 3, Δ_1 and Δ_2 are -0.6% and -0.4% respectively, thus satisfying the formula expressing the aforementioned structural conditions.

Regarding the optical characteristics, A_{eff} was 80 μm^2 or more and the cutoff wavelength was 1480 nm or less.

The transmission losses were 0.18 dB/km or less and 0.25 dB/km or less at wavelengths of 1550 nm and 1625 nm respectively and since no significant increase of loss was observed on the long wavelength side, it is considered that sufficient OCT is secured.

XT is -66 dB/km or less at a wavelength of 1625 nm and XT characteristics for transmission distances of over 10000 km were obtained in all communication wavelength bands. (Design Method)

FIG. 20 is a flowchart describing a design method for a multicore optical fiber.

The multicore optical fiber includes:

four cores having a radius a_1 , disposed in a longitudinal direction in square lattice shape;

a first cladding region disposed around the core and having a radius a_2 with a refractive index lower than the refractive index of the core, a specific relative refractive index difference between the cladding region and the core being Δ_1 ; and

a second cladding region disposed on an outer periphery of the first cladding region and having a refractive index higher than the refractive index of the first cladding region and lower than the refractive index of

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the core, a specific relative refractive index difference between the cladding region and the core being Δ_2 , in which

the design method executes:

a first step S01 of selecting a combination of the a_1 and the Δ_1 from requirements for effective cross-sectional area $A_{eff}(\text{m}^2)$ and a total value XT (dB/km) of inter-core crosstalk per km;

a second step S02 of selecting the a_2 and the Δ_2 that satisfy requirements for a cutoff wavelength and bending loss in the combination selected in the first step S01 and acquiring a core interval relationship between the Δ_1 with respect to the XT and a core interval Λ , and an OCT relationship between the Δ_1 and a shortest distance OCT from the outer periphery of the second cladding region to a center of the core with respect to requirements for an excessive loss α_c ; and

a third step S03 of calculating an outer diameter ϕ of the second cladding region at the Δ_1 acquired in the first step using the core interval relationship and the OCT relationship acquired in the second step S02 according to:

$$\phi = 2 \times (\Lambda \sqrt{2} + OCT)$$

and determining whether the outer diameter ϕ becomes 125 μm or less.

When a 4-core optical fiber having a standard cladding diameter of 125 μm is designed, requirements for the A_{eff} , the XT, the bending loss and the cutoff wavelength are set first (step S00).

Next, in step S01, a combination of a_1 and Δ_1 that satisfies the requirements for A_{eff} is selected as shown in FIG. 3 to FIG. 5. As shown in FIG. 10 and FIG. 11, a condition for Δ_1 corresponding to the desired XT is set. For example, when A_{eff} is 80 μm^2 or more, a combination of a_1 and Δ_1 that satisfies Formula 1 and Formula 3 is selected in first step S01.

Next, in step S02, as shown, for example, in FIG. 6, FIG. 7 to FIG. 9, a_2 and Δ_2 that satisfy the requirements for the bending loss and XT are selected for a_1 and Δ_1 selected in advance. More specifically, a_2 is selected from the graph in FIG. 12 obtained from FIG. 7 to FIG. 9 and Δ_2 is selected from the graphs in FIG. 13 and FIG. 14. At this time, parameters of the core 10 and the first cladding region 11 are obtained. For example, when a_2/a_1 is 3.0 or less, Δ_2 that satisfies Formula 3 is selected in second step S02. Since all a_1 , a_2 , Δ_1 and Δ_2 are available at this time, using FIG. 6, the requirements for XT, and a core interval and an OCT that can sufficiently reduce the excessive loss are calculated and a necessary cladding diameter is calculated.

Finally, in step S03, if the necessary cladding diameter is 125 ± 1 μm or less, the design is completed. If the cladding diameter is larger than 125 ± 1 μm , parameters of the core and the first cladding region are re-selected so as to obtain 125 ± 1 μm or less (repeat from step S01).

(Optical Transmission Method)

FIG. 21 is a diagram illustrating an optical transmission method using the aforementioned multicore optical fiber. The optical transmission method according to the present invention is characterized in that the multicore optical fiber 15 of 1000 km or more is used as an optical transmission path, signal light having a wavelength of 1530 nm or more is transmitted from four transmitters 21 for each core 10 of the multicore optical fiber 15, and four receivers 22 receive the signal light for each core 10 of the multicore optical fiber 15.

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The optical fiber cable **100** is provided with the 4-core optical fiber **15**, the transmitter **21** and the receiver **22** are connected to the transmitting side and the receiving side of each core **10** respectively. As described above, the optical fiber cable **100** has a maximum crosstalk of -54 dB/km or less and is suitable for long-distance transmission of several thousands of kilometers. Here, it is assumed that cable installation includes connection points between fibers and light amplifiers, but crosstalk in the fibers becomes dominant in transmission paths of several thousands of kilometers and influences of crosstalk at other connection points or optical components are considered to be sufficiently small.

The present optical transmission method can replace the single mode optical fiber of the submarine communication system having a communication distance of 1000 km or more with the multicore optical fiber **15**, transmit signal light having a wavelength of 1530 nm or more from the four transmitters **21** for each core **10** of the multicore optical fiber **15** and receive the signal light by the four receivers **22** from each core **10** of the multicore optical fiber **15**.

Since the multicore optical fiber **15** adopting a standard cladding diameter is used for the optical fiber cable, the existing optical cable structure can be reused, which is preferable.

INDUSTRIAL APPLICABILITY

The present invention can be used for optical fibers in optical communication systems.

REFERENCE SIGNS LIST

- 10** core
- 11** first cladding region
- 12** second cladding region
- 21** transmitter
- 22** receiver
- 100** optical cable

The invention claimed is:

1. A multicore optical fiber having only four cores having a radius a_1 disposed in a longitudinal direction in square lattice shape, wherein the multicore optical fiber further comprises:

first cladding regions disposed around each of the cores and having a radius a_2 with a refractive index lower than a refractive index of the cores, a specific relative refractive index difference to the cores Δ_1 ; and a second cladding region disposed on an outer periphery of the first cladding regions and having a refractive index higher than the refractive index of the first cladding regions and lower than the refractive index of the cores, a specific relative refractive index difference to the core being Δ_2 ,

wherein an outer diameter of the second cladding region is 125 ± 1 μm , bending loss at a wavelength of 1625 nm and a bending radius of 30 mm is 0.1 dB/100 turns or less, and a cutoff wavelength is 1530 nm or less.

2. The multicore optical fiber according to claim 1, further comprising a region having a width w and having a same refractive index as the refractive index of the second cladding region between each of the cores and the first cladding region.

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3. The multicore optical fiber according to claim 2, wherein the width w is $0 \mu\text{m} < w \leq 1.3 \mu\text{m}$.

4. The multicore optical fiber according to claim 1, wherein a relationship between the Δ_1 and a total value XT (dB/km) of inter-core crosstalk per km satisfies Formula C1,

[Formula C1]

$$\Delta_1 \leq 4.93 \times 10^{-5} XT - 0.00127 \quad (C1).$$

5. The multicore optical fiber according to claim 1, wherein a relationship between the a_1 and the Δ_1 satisfies Formula C2,

[Formula C2]

$$a_1 \geq -28041 \Delta_1^2 - 560.65 \Delta_1 + 3.1369 \quad (C2).$$

6. The multicore optical fiber according to claim 1, wherein a relationship between the Δ_2 and a total value XT (dB/km) of inter-core crosstalk per km satisfies Formula C3,

[Formula C3]

$$\Delta_2 \geq -1.14 \times 10^{-5} XT - 0.00509 \quad (C3).$$

7. The multicore optical fiber according to claim 1, wherein

a ratio a_2/a_1 between the a_1 and the a_2 is 1.0 or more and 3.0 or less,

a relationship between the Δ_1 , the a_2/a_1 , and an effective cross-sectional area A_{eff} at a wavelength of 1550 nm satisfies Formula C4 and

a relationship between the Δ_2 , the a_2/a_1 , and the A_{eff} satisfies Formula C5,

[Formula C4]

$$\Delta_1 \leq (1.9 \times 10^{-5} A_{eff}^2 - 0.00033 A_{eff} + 0.0163) a_2/a_1 + (-8.0 \times 10^{-6} A_{eff}^2 + 0.00139 A_{eff} - 0.0705) \quad (C4)$$

[Formula C5]

$$\Delta_2 \leq (5.0 \times 10^{-7} A_{eff}^2 + 6.5 \times 10^{-6} A_{eff} - 0.00108) a_2/a_1 + (-2.9 \times 10^{-7} A_{eff}^2 + 5.58 \times 10^{-5} A_{eff} - 0.00594)$$

and

$$\Delta_2 \leq (5.0 \times 10^{-7} A_{eff}^2 - 9.5 \times 10^{-5} A_{eff} + 0.0056) a_2/a_1 + (-1.6 \times 10^{-6} A_{eff}^2 + 0.00031 A_{eff} - 0.0208) \quad (C5).$$

8. The multicore optical fiber according to claim 5, wherein the Δ_1 and the Δ_2 satisfy Formula C6,

[Formula C6]

$$-0.7\% \leq \Delta_1 \leq -0.49\%$$

$$-0.4\% \leq \Delta_2 \leq -0.27\% \quad (C6).$$

9. The multicore optical fiber according to claim 7, wherein the Δ_1 and the Δ_2 satisfy Formula C6,

[Formula C6]

$$-0.7\% \leq \Delta_1 \leq -0.49\%$$

$$-0.4\% \leq \Delta_2 \leq -0.27\% \quad (C6).$$

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